Mammalian Vestibular Physiology

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Biophysics of the Peripheral End Organs

A. PRINCIPLES OF INERTIA-DEPENDENT SENSATION

The structural and functional division of the labyrinthine inertia-sensing organs into semicircular canals and otolith organs should be considered in conjunction with the fundamental fact that in three-dimensional space there are two sets of independent degrees of freedom of movement, namely three in rotation (angular movement) and three in translation (linear movement and gravitational acceleration). The physical layout of the semicircular canals is appropriate for transducing the three *angular* degrees of freedom of movement, *independently* both of end-organ location in the skull and of coincidental translational motion. This, together with their anatomical structure, leaves the otolith organs as prime contenders for inertial transduction of *linear* motion and gravity. However, confusion often arises here because, depending explicitly on their fixed location on the moving platform, a set of linear accelerometers can, in some circumstances, yield useful information about angular motion (Benson and Barnes, 1973). In due course we shall also see that there are interactions in the CNS between canal and otolith signals (Chapter 5). Before proceeding, therefore, it may be helpful to attempt clarification of some basic attributes of the two component parts of the peripheral mechanism.

Perhaps the essential physical principles involved can best be approached by drawing an analogy with inertial movement sensors employed in the guidance of flight and space vehicles (Barlow, 1964; Mayne, 1974). In one commonly used inertial guidance system, the three rotational and three translational degrees of freedom are respectively

transduced by sets of three orthogonally arranged gyroscopes and three orthogonally arranged linear accelerometers. To avoid the "confusing" influences of radial and tangential linear accelerations otherwise incurred by angular movement of the vehicle, these sensors are all mounted on a separate platform which is rotationally stabilized relative to space by means of the gyroscopes. In other words, the maneuvering vehicle itself rotates around the gyrostabilized platform which carries the linear accelerometers. Consequently, the attached linear accelerometers transduce purely linear motion in space, and the separate functions of the gyroscopes (canals) and linear accelerometers (otolith organs) become conceptually clear.

In the biological equivalent there is of course freedom to turn the head relative to space, although in view of the above analogy it may be of more than passing interest that highly effective reflex stabilization of angular head movements can be invoked (Borries, 1920; Magnus, 1924, 1925, 1926; Dunlap and Mowrer, 1930; Huizinga and van der Meulen, 1951; Suzuki and Cohen, 1964; Cohen et al., 1966; Jones and Milsum, 1965; Outerbridge, 1969; Outerbridge and Melvill Jones, 1971) through simple and complex vestibulocollic reflex pathways (Wilson and Maeda, 1974; Ezure et al., 1976; see Chapter 7) sometimes referred to as "cristospinal reflexes" (Henriksson et al., 1962). That such stabilization depends largely on the rotation-sensing canals seems clear from the fact that functional inactivation of parallel pairs of canals by mechanical obstruction of fluid circulation leads to a permanent tendency for the development of oscillatory head movements in the planes of those canals (Money and Scott, 1962; Watt, 1976b). When such movement takes place, either because of lack of reflex stabilization or because of normal voluntary head movements, the otolith organs are exposed to radial and tangential linear accelerations and these in turn combine with available canal signals in formulating the overall pattern of afferent information recognized as a meaningful movement by the CNS. However, as mentioned above, multimodal sensory interactions of the kind normally encountered in natural movement are considered later, and at this stage we should examine separately the physiological attributes of the two main components of the end-organ complex, namely the semicircular canals and the otolith organs.

B. THE SEMICIRCULAR CANALS

Our present understanding of canal mechanics depends on a number of key discoveries and conceptual developments. First, the early observations that surgical interference produces serious disorganization of pos-

ture and movement identified these organs as important sensory structures in their own right. Further, behavioral responses to the artificial movement of fluid in the canal lumen pointed directly to relative fluid flow within the canal as the likely source of relevant neural stimulation (reviewed by Ewald, 1892; von Cyon, 1908; Camis, 1930; see also references cited in Chapter 1). In turn the important question arose, could head movements cause fluid movement within the canals? The question was ripe for investigation by physical analysis using Newtonian principles, and hence it attracted the classical physicists' attention.

Prominent among these was the renowned Austrian natural philosopher, Ernst Mach, a strong exponent of the so-called positivist view which held that in natural science no statement can be accepted unless subject to empirical verification. Mach maintained that all such verification must in the end be reducible to "statements about sensations." The mechanical nature of hydrodynamic components in the canal promised well to serve his current objective, namely to obtain empirical verification at the level of the primary sense organs (Mach, 1875, 1886; see also Henn and Young, 1975). With introduction of the physicist came the need, as indeed is often the case today, to distill out an acceptable and relevant simplification of the real thing; in short, to assume a suitable physical model susceptible to meaningful interrogation by contemporary analytical methods. For this the important step forward was the realization that, rather than being semicircular, the endolymphatic canal comprises a functionally *closed* fluid circuit. We may take this as the starting point of our analysis of the mechanical end-organ response.

1. Basic Hydrodynamics

Consider a thin circular tube as in Fig. 3-1 with an unimpeded lumen, containing a known fluid of density ρ and viscosity η, and having a radius of curvature R and internal radius of the tube r. Now assume that the whole system begins to rotate from a stationary position at time t_0 in a counterclockwise direction, with an angular acceleration of $\ddot{q}$ (using the usual notation that, e.g., $\dot{x} \equiv dx/dt$ and $\ddot{x} \equiv d^2x/dt^2$). After a time lapse, t, the solid tube has turned through an angle q. But because of its mass-dependent inertia, the fluid tends to lag the tube so that a free-floating (frictionless) piston would have moved through a smaller angle (p) than the tube itself (q), as shown in Fig. 3-1. The angular acceleration of the fluid relative to space is therefore given by $\ddot{p}$, and hence a clockwise moment is generated due to the inertial forces acting on the fluid, given by the expression $I\ddot{p}$, where I is the moment of inertia of the fluid in the circular tube.

This moment leads to the generation of relative fluid flow in the tube

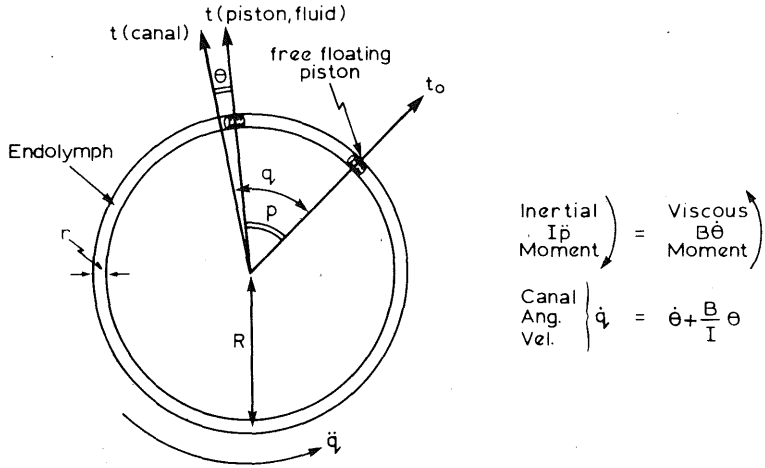

Fig. 3-1. Simplified circular canal *without* inclusion of the cupula, to illustrate the fundamental "integrating" function of canal hydrodynamics. $q, \dot{q}, \ddot{q}$, Angular position, velocity, and acceleration of the canal in its own plane; t_o, location of "indicator" piston at start of counterclockwise rotation; t (canal), t (fluid), locations of canal and fluid at time t after initiation of counterclockwise angular acceleration; p, absolute angle of fluid movement relative to space in time t; r, internal radius (n.b., not diameter) of canal; R, radius of curvature of the circular tube; I, moment of inertia of fluid contained in the tube; B, moment of viscous friction per relative angular velocity ($\dot{\theta}$) of fluid flow.

and hence to an opposing moment due to viscous forces. *Provided that the pattern of this flow is strictly laminar* and assuming steady flow, then with $r \ll R$ this moment due to viscous forces must be strictly proportional to the *rate* of *relative* flow, expressed as $\dot{\theta}$, where θ denotes the *relative* angular displacement of fluid around the circuit of the canal.

Since in Newtonian flow such as this no other forces act on the fluid, the net moments in opposite directions must equal one another, so that

$$I\ddot{p} = B\,\dot{\theta} \tag{3.1}$$

where B is the moment of viscous friction per relative angular velocity of fluid flow around the canal. But since $p = q - \theta$, and assuming linearity, then

$$\ddot{p} = \ddot{q} - \ddot{\theta} \tag{3.2}$$

and hence, substituting in equation (3.1) for $\ddot{p}$,

$$I\,(\ddot{q} - \ddot{\theta}) = B\dot{\theta} \tag{3.3}$$

Alternatively,

$$\ddot{q} = \ddot{\theta} + \frac{B}{I}\dot{\theta} \tag{3.4}$$

Integrating with respect to time and assuming zero initial conditions,

$$\dot{q} = \dot{\theta} + \frac{B}{I}\theta \tag{3.5}$$

Hence, for a step velocity input of amplitude $\dot{Q}$,

$$\theta = \frac{I}{B} \cdot \dot{Q}\,(1 - e^{-(B/I)\,t}) \tag{3.6}$$

Therefore, when t is large relative to the exponential time constant I/B [see equation (3.8)], $e^{-(B/I)\,t}$ tends to zero and

$$\theta \simeq \frac{I}{B} \cdot \dot{Q} \tag{3.7a}$$

Several questions arise. First, how strictly laminar is the fluid flow in the canal? The answer to this is to be found in the very small Reynolds number of the system, which being less than 1 is far below the critical value associated with transition from laminar to turbulent flow. Consequently, the flow must be strictly laminar and hence the hydrodynamic integration should be accurate.

Second, how long must we wait for a reasonably complete transition from equation (3.6) to equation (3.7a)? Here we must look to the exponential term in equation (3.6), where the value of the constant B/I has the dimension 1/time, so that its inverse emerges as a time constant, T_1, defined numerically by the relation

$$T_1 = \frac{I}{B} \tag{3.8}$$

In other words, T_1 is the time taken for the fluid displacement to reach $1/e$ of the asymptotic value, θ, determined by the step change of angular velocity of the canal, $\dot{Q}$, as in Fig. 3-2. Here the lower curve depicts a step change in canal angular *velocity*, while the upper one gives the resulting exponential form of relative fluid *displacement* in the canal. If T_1 is short compared to most head movements, then the relative fluid displacement becomes, for all practical purposes, coded as instantaneous head velocity. If not, then we have to seek further for a meaningful interpretation of response. To settle the matter, we have to evaluate the constants I and B numerically.

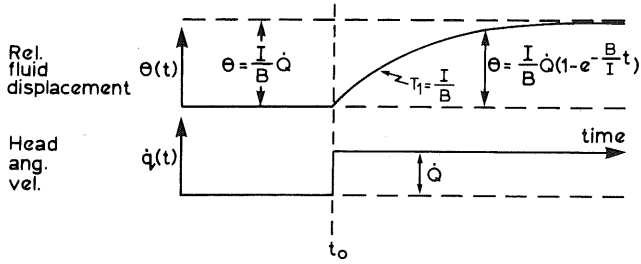

Fig. 3-2. Hydrodynamic response (θ) of the simple circular canal of Fig. 3-1 to a step change of angular velocity relative to space. A sudden change of head angular velocity ($\dot{Q}$) at t_0 leads to an exponential pattern of relative fluid displacement reaching an asymptotic value of $I/B \cdot \dot{Q}$ with a time constant $T_1 = I/B$.

Various solutions are available in the literature (von Gaede, 1922; Schmaltz, 1925, 1932; Steinhausen, 1933; van Egmond et al., 1949; Mayne, 1950; Jones and Milsum, 1965; Fernández and Valentinuzzi, 1968; Melvill Jones and Milsum, 1971; Money et al., 1971; Oman and Young, 1972; Fernández and Goldberg, 1971; Curthoys et al., 1977b). Rather than catalogue the results of these several authors, we will derive approximate values from first principles in order both to illustrate how this can be done for particular circumstances and to examine the physiological significance of the general outcome.

First, the moment of inertia, I, of a mass, m, of distance l from the axis of rotation is given by the relation

$$I = ml^2 \tag{3.9}$$

In the circular tube of Fig. 3-1 the relevant mass of fluid is given by the product of its density, ρ, and its volume, the latter being given by the product of the internal cross-sectional area of the tube, πr^2, and its circumference, $2\pi R$. Thus substituting $\rho \cdot \pi r^2 \cdot 2\pi R$ for m and R^2 for l^2 in equation (3.9), we have for the contained fluid, assuming $n \ll R$,

$$\begin{aligned} I &= \rho\pi^2 r^2 \cdot 2R \cdot R^2 \\ &= 2\rho\pi^2 r^2 R^3 \end{aligned} \tag{3.10}$$

This expression contains measurable variables so that an approximate value for I can be estimated knowing only the density of endolymph, the mean radius of curvature of the canal, and the internal radius of the endolymphatic tube, provided that a consistent system of units is em-

ployed. In practice, however, it is the ratio I/B which we are seeking and hence we must next derive an expression from which to calculate B.

For this we may begin by stating the Poiseuille-Hagen formula describing viscous flow in thin tubes, where the flow is laminar and, in steady state, takes a parabolic profile of linear flow velocity across the diameter of the tube.

In these circumstances the net frictional force, F, due to inertially generated fluid flow is given by

$$F = \frac{8\bar{V}\eta l}{r^2}$$

where $\bar{V}$ is volume flow per second, η is viscosity of the fluid, and l is the circumferential length of the tube; since in this case $l = 2\pi R$,

$$F = \frac{8\bar{V}\eta \cdot 2\pi R}{r^2}$$

$$= \frac{16\bar{V}\eta \cdot \pi R}{r^2}$$

The turning moment, M, of this force acting at radius R is therefore given by

$$M = F \cdot R = \frac{16\bar{V}\eta \cdot \pi R^2}{r^2}$$

But the coefficient B is defined as the turning moment M due to frictional force F per unit angular velocity of fluid flow, $\Omega = \dot{\theta}$. Hence, since the average relative circumferential linear velocity, v, of fluid flow is given by

$$v = \frac{\bar{V}}{\pi r^2}$$

where $\bar{V}$ represents the volumetric flow rate, and the average angular velocity of relative flow, Ω, around the circle of radius R by

$$\Omega = \frac{v}{R} = \frac{\bar{V}}{\pi r^2 \cdot R}$$

then

$$B = \frac{M}{\Omega} = \frac{16\bar{V}\eta \cdot \pi R^2}{r^2} \cdot \frac{\pi r^2 R}{\bar{V}}$$

$$= 16\eta\pi^2 R^3 \tag{3.11}$$

.0163 cm

Combining equations (3.10) and (3.11),

$$\frac{I}{B} = \frac{2\rho\pi^2 r^2 R^3}{16\pi^2\eta R^3} = \boxed{\frac{\rho r^2}{8\eta}} \tag{3.12}$$

In this expression ρ and η have been determined experimentally by a number of investigators (e.g., Money et al., 1966, 1971; Steer, 1967; ten Kate et al., 1970; see Table 2-1 for reported values) yielding values close to 1.0 and 0.01, respectively, expressed in the centimeter-gram-second (cgs) system of units. In man 0.0163 cm has been quoted by Curthoys et al. (1977b) as a likely mean value for the internal radius of the living endolymphatic tube.

Then

$$\frac{I}{B} = \frac{1.0 \times (0.0163)^2}{8 \times 0.01} \text{ sec}$$

$$= 0.003 \text{ sec} \tag{3.14}$$

Summarizing, this very small value of I/B tells us that if the simple model of a circular tube is plausible, then in man the integrating hydrodynamic response seen graphically in Fig. 3-2 would be so rapid compared to the usual durations of head movement that the angular velocity-dependent response could for practical purposes be considered instantaneous, and the approximation of equation (3.7a) be written as an equality.

Another important feature of equation (3.7a) is that the same term I/B can be seen to constitute the numerical gain, G, of the response, expressed as the relative angular displacement of contained fluid, θ, per angular velocity of the canal, $\dot{Q}$. Thus

$$G = \frac{I}{B} = \frac{\theta}{\dot{Q}} \tag{3.7b}$$

Consequently, the above numerical value of $I/B = 0.003$ sec determines a response sensitivity on the order of 3×10^{-3} deg of fluid displacement per deg/sec of head angular velocity. (Note that it is quite in order for this value to have the dimensions of time, since so does the gain, G, expressed as deg per deg/sec). In cat, where the value of r has been quoted as 0.0115 cm (Fernández and Valentinuzzi, 1968), the corresponding sensitivity would be less than this, namely about 2×10^{-3} deg per deg/sec.

Thus an important general conclusion is that even very high head angular velocities of, say, 500°/sec (Hallpike and Hood, 1953) would incur a fluid circulation only on the order of 1°. With human subjective thresholds of around 2°/sec (van Egmond et al., 1949), the implication would be that the end organ is endowed with exquisitely sensitive mechanoneural transduction such that no more than about 6/1000th of a degree of fluid circulation around the canal would generate a detectable neural response,

although this may represent a higher sensitivity than occurs in the real canal (Curthoys et al., 1977b) for reasons discussed later.

Early theorists, e.g., Mach, von Gaede, and Schmaltz, were aware of the small order of magnitude of this physical constant and also of the two important conclusions which must follow, namely that (1) the actual fluid flow due to head rotation would be exceedingly small and (2) the integrating response would be virtually instantaneous for normal head movements. Indeed, the inferred fluid flow was so small as to call into question the presumption that it constitutes the adequate stimulus for the neural response (Goltz, 1870).

Before leaving the elementary rigid circular model of Fig. 3-1, one further fundamental feature of the simple "toroid" canal should be mentioned, namely that its hydrodynamic response is independent of the position of the axis of rotation (e.g., von Gaede, 1922). In other words, wherever the canal may be located in the skull, the *net* inertial force causing relative fluid flow around the circular tube of the canal depends only on its angular acceleration and not at all on any form of translational acceleration, at least for the idealized torus of Fig. 3-1.

The simple circular model serves to provide considerable physical insight, beyond the reach of intuition, into how such a system would respond to components of angular movement in its own plane. The next question is, to what extent can these insights be applied to the real biological equivalent?

2. Mechanical Significance of the Cupula

For many years the existence of the cupula membrane was questioned, largely on account of its liability to dislocation and removal by histological and staining techniques of the time. It remained for the imaginative and technically adept demonstration of a moving ampullary cupula in living preparations (Steinhausen, 1931, 1933; Dohlman, 1935) to enable investigation of its likely influence on the hydrodynamics of the simple model of Fig. 3-1.

Figure 3-3 replicates this model, *with the addition of a linear spring component*, always tending to force the "piston" back to a datum or "zero" position in the canal, with a force proportional to its displacement from that position. This addition of a simplified "cupula" to the model introduces three significant new features. First, physical movements of living structures must be contained within anatomical limits of the actual cupula-ampullary system. That this does not impose a problem is predicted with fair certainty by the very small gain of the simple model,

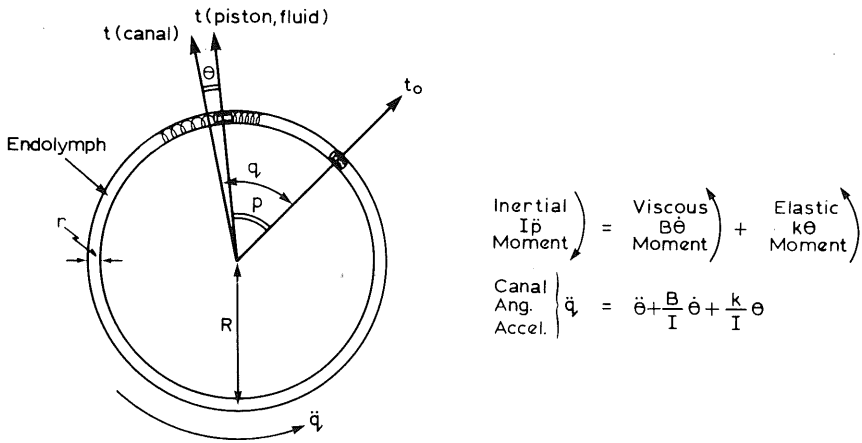

Fig. 3-3. As in Fig. 3-1, with the introduction of a spring restoring force acting on the piston, simulating a watertight, elastic cupula. K, Spring restoring force per relative angular displacement of fluid around the tube. Other symbols as in Fig. 3-1.

calling as it does for a mere 2/1000 to 3/1000 deg (i.e., about 1 sec of arc!) of fluid circulation around the tube per deg/sec of change in head angular velocity. Second, the spring characteristic will prevent dc drifting of the mechanical datum of the sensor, ensuring that in the absence of angular acceleration the "cupula" will always be restored eventually to the same zero position. Third, a new force is introduced into the system which, being position dependent, turns the first-order equation of the simple torus [equation (3.1)] into the following second-order equation by adding the term, $K\theta$, so that

$$I\ddot{p} = B\dot{\theta} + K\theta \tag{3.15}$$

where K is the spring restoring force per unit angular displacement of fluid around the circuit. As before, since $p = (q - \theta)$

$$\ddot{q} = \ddot{\theta} + \frac{B}{I}\dot{\theta} + \frac{K}{I}\theta \tag{3.16}$$

The very significant functional implications of this apparently trivial development are not intuitively obvious and require further analytical treatment for their elucidation. For this purpose we will convert the above differential equation to the corresponding Laplace transfer function, since application of the engineering method of systems analysis then facilitates

conceptual interpretation of physiologically meaningful consequences. Thus in practice the *general* form of response of a given transfer function can readily be looked up (e.g., Milsum, 1966), and, from this, detailed numerical solutions for all situations are readily available. Furthermore, the analytical formulation of interactions between concurrently active dynamic terms in different parts of an informational chain is greatly facilitated.

Moving from the conventional time domain of equation (3.16) to the Laplace frequency domain,

$$\ddot{q}\,(s) = \theta\,(s)\left(s^2 + \frac{B}{I}s + \frac{K}{I}\right)$$

where $\ddot{q}(s)$ and $\theta(s)$ denote the Laplace transforms of $\ddot{q}(t)$ and $\theta(t)$. A great practical advantage of this step is that differential equations in the Laplace frequency domain may, with certain limitations, be treated as algebraic equations. Thus, rearranging,

$$\frac{\theta}{\ddot{q}}(s) = \frac{1}{s^2 + (B/I)s + (K/I)}$$

Multiplying numerator and denominator by I/K,

$$\frac{\theta}{\ddot{q}}(s) = \frac{I}{K} \cdot \frac{1}{(I/K)s^2 + (B/K)s + 1}$$

and factorizing the denominator of the right-hand side,

$$\frac{\theta}{\ddot{q}}(s) = \frac{I}{K} \cdot \frac{1}{(T_1 s + 1)\,(T_2 s + 1)} \tag{3.17}$$

where $T_1T_2 = I/K$ and $T_1 + T_2 = B/K$.

An important step forward was made when van Egmond et al. (1949) showed experimentally that the value of T_2 is much larger than the "short" time constant of equation (3.14), $T_1 = I/B$, so that the value $T_1 + T_2 = B/K$ becomes a close approximation to the "long" time constant, T_2. To a close approximation, therefore, we may write

$$T_2 = B/K \tag{3.18}$$

As a result, we may also write, to a close approximation,

$$T_1T_2 = T_1 \cdot B/K = I/K$$

and hence

$$T_1 = I/B \tag{3.19}$$

The transfer function of equation (3.17) then becomes

$$\frac{\theta}{\ddot{q}}(s) = \frac{I}{K} \cdot \frac{1}{[(I/B)s + 1][(B/K)s + 1]} \tag{3.20}$$

This describes the relevant output (i.e., relative fluid displacement, θ) as a function of the input which is head angular acceleration, $\ddot{q}$. Taking advantage of the earlier conclusion (equation 3.7a) that the response is likely to be most meaningfully related to instantaneous angular velocity, we may make the substitution $\dot{q}s = \ddot{q}$ and so arrive at the relation

$$\frac{\theta}{\dot{q}}(s) = \frac{I}{K} \cdot \frac{s}{[(I/B)s + 1][(B/K)s + 1]} \tag{3.21a}$$

In the present context it is instructive to examine the frequency response of the system represented by this transfer function; that is, the phase and gain relationships of the input ($\dot{q}$ = head angular velocity) and output (θ = "cupula" displacement) over a wide range of sinusoidal frequencies of stimulation, say from 0.01 to 10.0 Hz. The form of this frequency response is shown in Fig. 3-4. The classical Bode plot is completed by extrapolation of the (dashed) straight lines, intersecting one another at the so-called asymptotic cutoff frequencies designated

$$\omega_1 = \frac{1}{T_1} = \frac{B}{I} \qquad \text{and} \qquad \omega_2 = \frac{1}{T_2} = \frac{K}{B} \tag{3.22}$$

where ω denotes frequency in radians/sec ($\omega + 2\pi \times$ Hz).

When interpreting this figure it is important to note that the forcing function has been expressed in terms of angular velocity, $\dot{q}$. The "flat" middle range of response, with relatively constant gain and minimum change of phase with frequency, therefore describes a frequency range over which the instrument will deliver a "cupula" displacement that is a close analogue of "instantaneous" angular velocity of the canal. Bearing in mind the velocity-transducing character of the simplest model (Fig. 3-1), one is led to suspect that this range may correspond to that of natural movement. To examine this possibility we need to estimate likely values of the frequencies ω_1 and ω_2 which bracket this "middle-frequency regime." We already have a value for I/B of 0.003 sec from which we can predict the *upper* cutoff frequency as 1/0.003 = 333 rad/sec or around

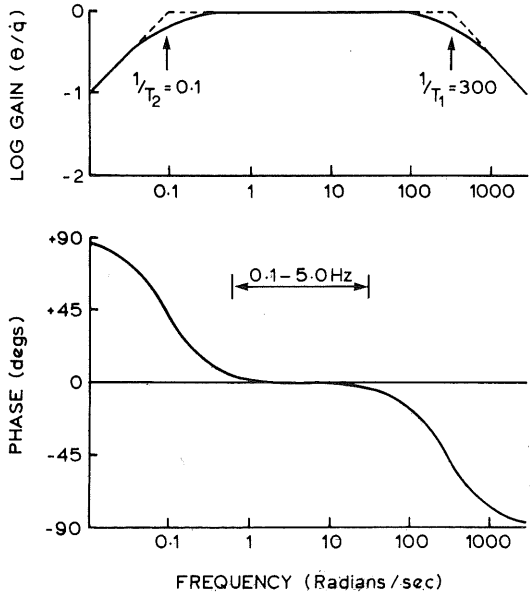

Fig. 3-4. Bode plot of the frequency response of equation (3.21b), representing the dynamic response of the simplified "cupula" canal model of Fig. 3-3. Note that both gain and phase are expressed with respect to head angular velocity ($\dot{q}$) · $T_1 = I/B$, $T_2 = B/K$. (After Jones and Milsum, 1965.)

50 Hz. Excepting vibrational stimuli, this clearly is well above the range of normally occurring smooth head movement.

The lower cutoff frequency cannot be so readily calculated since, although as already seen [equation (3.11)] the coefficient B can be estimated analytically, the same does not apply to the "cupula" elasticity coefficient K, which must be determined experimentally.

Van Egmond et al. (1949), using new methods of human experimentation, and Mayne (1950), using a systems engineering approach to interpret previously published data (e.g., Mowrer, 1932, 1935; Lowenstein and Sand, 1936, 1940), almost simultaneously and quite independently estimated the time constant B/K as at least 10 sec. From such estimates, and knowing B, it is then of course possible to calculate estimates for K. But from a functional viewpoint it is the time constant itself that determines the lower cutoff frequency, which, being $1/T_2$, thus turns out to be on the order of 0.1 rad/sec = 0.016 Hz ($\simeq$ 1/60 Hz) or lower. Again this value would seem to be well below the lower range of frequency content of

normal head movement, implying that in normal circumstances the canal is mainly subject to stimuli which are well within the middle-frequency range of Fig. 3-4.

Furthermore, within this "middle" range, the gain, G_m, of response expressed as relative angular displacement of the fluid, θ, per unit change in angular velocity of the canal relative to space can be shown to be essentially the same as that for the simple torus, namely I/B (e.g., Melvill Jones and Milsum, 1971). Thus, since in equation (3.21a), $I/K = T_1T_2$, we may rewrite this equation as

$$\frac{\theta}{\dot{q}}(s) = \frac{T_1T_2s}{(T_1s + 1)(T_2s + 1)} \tag{3.21b}$$

In order to convert from the frequency domain (s) to the (real) time domain (t), we replace s by $j\omega$, where $j = \sqrt{-1}$ and ω is, as before, the sinusoidal frequency of rotational stimulation expressed in rad/sec. Then, *for frequencies well within the "middle" range*, $\omega T_1 << 1$ and $\omega T_2 >> 1$ (bearing in mind that the upper and lower cutoff frequencies are $1/T_1$ and $1/T_2$, respectively, as in Fig. 3-4); hence

$$G_m = \frac{\theta}{\dot{q}}(t) = T_1 = \frac{I}{B} \tag{3.7b}$$

At this point it is worth noting that since at frequencies *well below that of the upper cutoff value* ($\omega_1 = 1/T_1 \simeq 300$ Hz) the product (T_1s) becomes much less than 1, the middle- to low-frequency response of the system is adequately defined by the simplified relation

$$\frac{\theta}{\dot{q}}(s) = \frac{T_1T_2s}{T_2s + 1} = \text{const.}\ \frac{s}{T_2s + 1} \tag{3.21c}$$

This simplified equation will be used later (Chapters 4 and 5) to aid interpretation of neurophysiological responses to natural stimuli.

Summarizing, we may conclude that in all probability introduction of a weakly elastic cupula into the canal system does not in practice modify the main inference drawn from analysis of the simple torus with unimpeded hydrodynamics. In effect, the "design mode" of the instrument still appears to be that of an integrating angular accelerometer yielding a mechanical response which is an analogue of instantaneous angular velocity, with the new constraint, however, that this conclusion applies only over a limited bandwidth of frequencies which would seem likely to include the range of most naturally occurring head movement. Seen in this context, the "cupula" in practice simply adds a means of converting the hydrodynamic response into tissue movement suitable for neural transduction of the "sensed" rotation, with the added advantage that its elastic

"spring" properties ensure permanent maintenance of a fixed cupula datum representing zero rotation.

3. Allometry[1] and the Functional Significance of Canal Size

The above biophysical considerations lead to a conceptual interpretation in which, specifically over a "middle" range of sinusoidal frequencies, the system, acting as an integrating angular accelerometer, produces a response whose informational content is consistently proportional to instantaneous head angular velocity. Since, at least for man, this "middle" range probably covers a wide range of natural head movement, it seems reasonable to speculate that evolutionary selection has in some way bespoken the need for this specifically velocity mode of transduction of angular head movement. The speculation finds support from a recurrent observation that in sophisticated man-controlled vehicles, such as helicopters and high-performance aircraft, it proves advantageous to design instrument attitudinal displays which provide the human sensory input with considerably greater "velocity" than "positional" information about the vehicle's movement.

Given this lead, together with the fact that convenient limits to the "middle," or velocity-transducing, frequency range are bound by the asymptotic "turnover" frequencies in Fig. 3-4, one could speculate further that these limits should move up or down the frequency scale according to the frequency content of normal head movements in different animals and habitats. Small animals with short-sharp head movements would require these limits to move up the frequency scale; in large animals with more ponderous head movements the limits should move down the scale.

It turns out that since, as described above [equations (3.18) and (3.19)], the lower and upper "turnover" frequencies are defined by $1/T_2 = K/B$ and $1/T_1 = B/I$, respectively, the hypothesis may be tested numerically, given the physical dimensions of the canals, a value for K, and some criterion for estimating the dependence of head movements upon animal size.

Using the method of dimensional analysis (e.g., Bridgman, 1931; Taylor, 1974) and the assumption that stress (i.e., force per unit area) in the neck due to head weight is independent of animal size, Jones and Spells (1963) drew the conclusion that, for similarity in the different time scales associated with the head movements of animals of different size,

[1]*Allometry* (Greek *allos* = other; *metron* = measure): study of relative growth; change of proportions with increase of size; growth rate of a part different from a standard growth rate or from the growth rate of the whole (from Kenneth, 1963).

the radius of curvature (R) and the square of the internal radius of the lumen (r^2) of the model canal should vary as the same power function of body mass (M^n). Thus

$$R \propto r^2 \propto M^n \tag{3.23}$$

To test this prediction, these authors plotted measured values of r^2 and R against animal mass for 46 different mammalian species, using log-log scales to determine the anticipated power relationship. Most of these measurements were made from Gray's unique collection of decalcified labyrinths located in the Royal College of Surgeons, London, and recorded in two remarkable volumes of scaled, stereoscopic photographs (Gray, 1907, 1908). The results are shown in Fig. 3-5A,B. Linear log-log

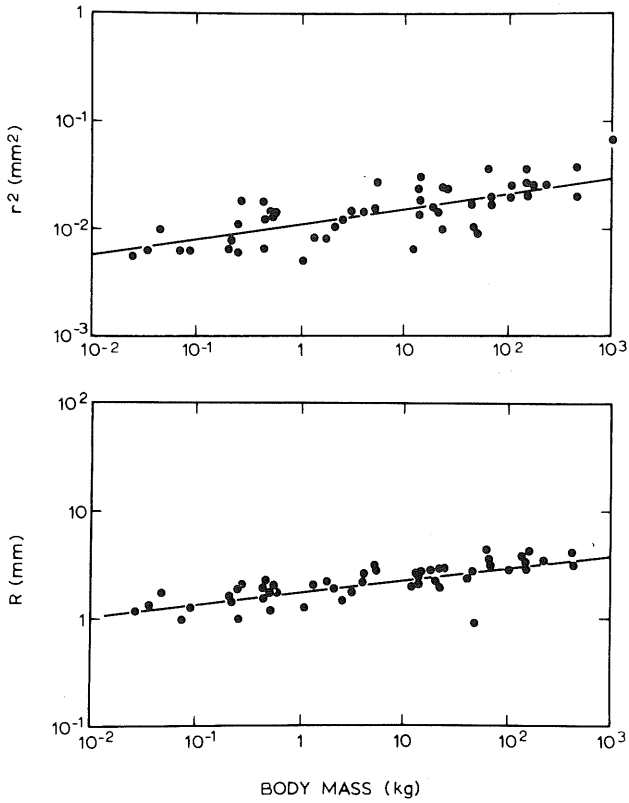

Fig. 3-5. Dependence of critical canal dimensions on size of animal in 46 mammalian species. r, Internal radius of horizontal endolymphatic canal; R, radius of curvature of the semicircular canal. Note that both ordinates and abscissae are logarithmic. (Jones and Spells, 1963.)

regression lines are well fitted by the data points in each case, confirming the power relationship of equation (3.23). Furthermore, the slopes of the two lines are highly significant ($P < 0.001$) and statistically indistinguishable from one another ($P > 0.4$), the respective values of these slopes being 0.12 (SE 0.015) and 0.14 (SE 0.021). An extended investigation over a wider range of animals including nonmammalian species yielded corresponding values of 0.09 and 0.08, respectively. The value predicted on the basis of dimensional analysis was 0.083.

The conclusion was drawn that there are very small (note the log-log scales of Fig. 3-5) but highly consistent increases of canal dimensions with animal size and that these increases accord well with the prediction that they are roughly what would be required for maintenance of dynamic similarity of the canal's response throughout the range of mammalian species examined. The functional implication becomes clear when examined in the context of the system's frequency response.

As already seen, the values of I/B and B/K numerically determine the limits of the "middle," velocity-transducing, range of frequencies, as in Fig. 3-4. But as shown in equation (3-12)

$$I/B \propto r^2 \tag{3.24}$$

Also, since K has been given as (Jones and Spells, 1963)

$$K = \pi \mu r^2 R$$

where μ is the pressure exerted by the cupula per unit angular deflection of fluid round the circuit, then from equation (3.11)

$$\frac{B}{K} = \frac{16\pi^2\eta\, R^3}{\pi \mu r^2 R} \propto \frac{R^2}{r^2} \tag{3.25}$$

But from the above similarity criteria it has already been shown that $r^2 \propto R$, so that if these criteria are met then from equations (3.24) and (3.25)

$$\frac{B}{K} \propto R \propto r^2 \propto \frac{I}{B} \tag{3.26}$$

Summarizing, this conclusion predicts that with increasing animal size the two time constants $T_1 (\simeq I/B)$ and $T_2 (\simeq B/K)$ would increase proportionately to one another and also proportionately to the same function of body mass, M^n, as would r^2 and R [equation (3.23)] and hence also as R^2/r^2. The significant inference is that the "middle"-frequency bandwidth of the canal would move up and down the frequency scale by roughly that amount which would maintain similarity of canal response, i.e., specifi-

cally angular velocity transduction, over the predicted range of animal head movement according to animal size.

The close match between measured and predicted changes of physical canal dimensions seems highly suggestive that the velocity-transducing characteristic has in fact been favored by the evolutionary process. It seems unlikely that these systematic dimensional changes, which are very far indeed from bearing a simple proportionate relationship to animal size, could have occurred by chance over such a wide range of animals.

As detailed elsewhere (Melvill Jones, 1974), there are assumptions in the above argument. For example, it is tacitly assumed that the values of μ, ρ, and η do not vary systematically with animal size. Again the thickness of neck muscles is assumed to increase more than proportionately with animal size according to the specific assumption that stress in the neck due to head weight is maintained constant for all animals. This feature has certainly been noted in Thompson's remarkable study *On Growth and Form* (1942). However, the generalization might not hold for the erect primates, who do not require the otherwise expected muscular ability to hold the head up against gravity as do animals with a horizontal neck. No doubt other factors must be at play. For example, Mayne (1965) suggested that head movements might be constrained by the need to maintain inertial forces on brain tissue within acceptable limits. Indeed, he showed that the introduction of this factor would tend to increase the value of the exponent, n in equation (3.23), perhaps even beyond the value measured from the data in Fig. 3-5A,B.

Other investigators in this field have studied the significance of habitat, especially the effect of a fluid medium on the movements of fish (ten Kate, 1969; ten Kate et al., 1970; Howland and Masci, 1973a,b). Also, Howland and Masci (1973a,b) as did Jones and Spells (1963) noted that relative fixation of the fish's body with respect to the head must modify the allometry of the system, and indeed could account for the observed increase of canal dimensions relative to the body, which was noted in fish. Howland and Masci (1973b) even suggested a size-dependent increase of the cupula spring constant, μ, with body mass, M, of the form (again in fish)

$$\mu \propto M^{0.1} \tag{3.27}$$

Nevertheless, despite the many possible factors at play and an extensive interrogation of their physical and functional significance, Howland and Masci endorsed the conclusion that, taken as a whole, the data generally support the theory that the dimensions of the semicircular canals attune them to the angular frequency spectra of the animal's natural movement in its natural habitat.

4. Low-Frequency Response of the Canal System

The preceding sections lead to the conclusion that natural selection has evolved a canal system which in its normal functional role acts as an angular velocity transducer over a limited bandwidth of frequencies corresponding to those normally incurred during natural movement. In man, natural head movement during locomotion, stepping, and jumping extends up to at least 5 Hz (Benson and Gresty, personal communication, see also Fig. 8-20), and 0.1 Hz would seem a reasonable lower limit. From the frequency response of Fig. 3-4 it can be seen that between these limits the system would provide reliable velocity transduction (i.e., a "flat" response) with almost zero phase.

However, at low frequencies, in particular below the lower cutoff level, marked deviations from this characteristic occur. The "gain" of response decreases with a slope of 1 (i.e., 45°) on the log-log plot, and the phase becomes progressively advanced with respect to rotational velocity to reach eventually a value of 90° phase advancement, which would be in phase with angular acceleration. In these circumstances the response can best be described as that of a sluggish angular *accelerometer*, a feature which becomes apparent in Fig. 3-6 plotting the corresponding frequency response for the transfer function relating mechanical response, θ, to the *acceleration*, $\ddot{q}$, of the imposed rotational movement,

$$\frac{\theta}{\ddot{q}}(s) = \frac{I/K}{T_2 s + 1} \tag{3.21d}$$

Below the turnover frequency of $\omega = 1/T_2$ rad/sec, the *acceleration* gain becomes flat down to dc conditions, with the response phase registered as zero relative to acceleration (i.e., 90° phase advanced relative to velocity).

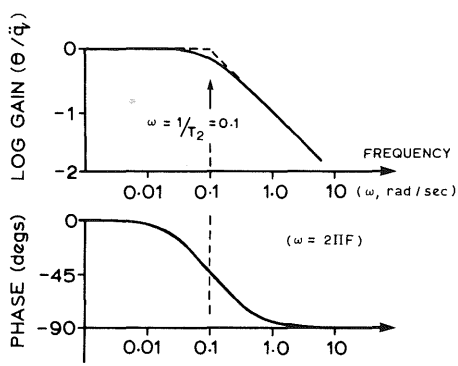

Fig. 3-6. Bode plot of the low-frequency response of equation (3.21d) expressed with respect to head angular acceleration ($\ddot{q}$).

An important feature of implied response characteristics is shown in Fig. 3-7, which gives the response of equation (3.21d) to a step change of angular acceleration or, what amounts to the same thing, a ramp change of angular velocity. Initially the hydrodynamic response, θ, is proportional to the increasing stimulus angular velocity. But with time it deviates exponentially from this relationship to approach an asymptotic value proportional to the ongoing acceleration, and this response is maintained indefinitely as long as the acceleration is maintained. The time constant of this exponential curve is $T_2 = B/K \simeq 10$ sec for man. Thus eventually, i.e., after 2–3 time constants duration of acceleration, or about 30–40 sec in the case of man, the cupula would have deviated through an angle closely proportional to the imposed acceleration, the gain of this (sluggish) low-frequency response (G_l) being expressed mathematically as (Melvill Jones and Milsum, 1971)

$$G_l = \frac{\theta}{\ddot{q}} = T_1 T_2 \tag{3.28}$$

$$= \frac{I}{B} \cdot \frac{B}{K} = \frac{I}{K} \tag{3.29}$$

Note incidentally that since $I/B = T_1 = G_m$ [equation (3.7b)], we may also write

$$G_l = G_m \cdot T_2 = G_m \cdot B/K \tag{3.30}$$

This simple relationship therefore allows one to compare results from different authors expressing sensitivity measurements as functions of angular acceleration (e.g., Goldberg and Fernández, 1971a) or of angular

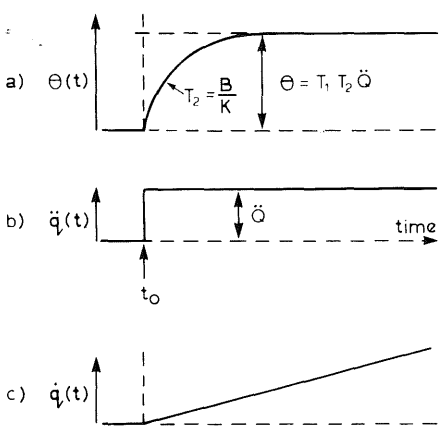

Fig. 3-7. (a) Response (θ) of the simplified canal model in Fig. 3-3 [equation (3.17)] to (b) a step change of angular *acceleration* or (c) a ramp change of angular velocity. A prolonged steady acceleration generates a slow, exponentially rising hydrodynamic response of time constant $T_2 = B/K$ and asymptotic value T_1T_2, when expressed as a function of the imposed acceleration. Note that this is not the same curve as that of Fig. 3-2, the effect of which is too short to manifest itself here.

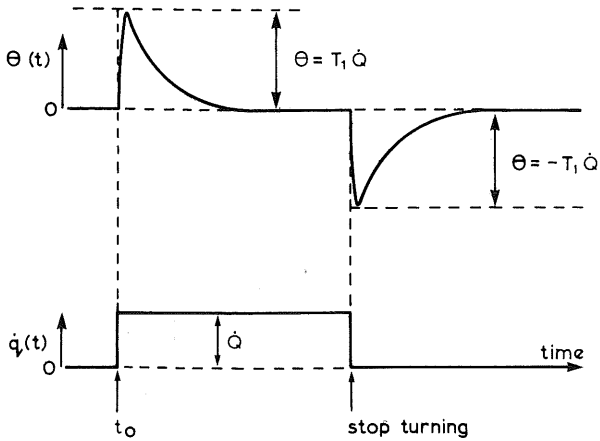

Fig. 3-8. Response (θ) of the simplified canal [Fig. 3-3, equation (3.21b)] to sudden initiation and cessation of a steady angular velocity, as employed in the classical Bárány turning test.

velocity (e.g., Melvill Jones and Milsum, 1971), provided of course that the value of T_2 is known.

Obviously the low-frequency response is of little significance in natural movement. But the phenomenon is often utilized in experimental work as a convenient means of establishing an exponential approach to a *steady*, maintained cupula deviation (Fig. 3-7) which cannot be achieved by any other means of adequate rotational stimulation.

Another unnatural pattern of stimulation, commonly used in clinical turning tests, comprises a stepwise change of maintained angular *velocity*. The idealized step response of equation (3.21b) is shown in Fig. 3-8. A sudden change of angular velocity from one steady value to another leads to a rapid velocity-dependent hydrodynamic response, which then decays exponentially to zero. Here again the relevant time constant is $T_2 = B/K \simeq 10$ sec. If after return to zero response the steady rotation is then suddenly stopped (Bárány's original clinical turning test, 1906, 1907), the associated deceleration will cause an equal, but opposite, initial response which again will decay to zero with the same exponential term as before. It may be noted that in reality the inertioviscous time constant $T_1 = I/B$ operates during the initial period of movement, slightly retarding the peak response, as indicated exaggeratedly in the figure. But, in practice, with a numerical value of I/B on the order of 0.003 sec, this effect would be negligible during most ordinary head movements.

Other low-frequency patterns of stimulation are encountered by man during the clinician's diagnostic procedure of thermal (caloric) stimulation

of the canal and the slow, prolonged turning maneuvers often experienced in flight and space and even the automobile, although these are not further considered in this monograph.

Summarizing, the response of the canal-cupula system to *low* sinusoidal frequencies of rotation, including stepwise dc changes of rotational velocity, is that of a sluggish angular accelerometer. This does not invalidate the earlier conclusion that in the middle band of frequencies depicted in Fig. 3-4 the mode of transduction is that of angular velocity since both conclusions are derived from the same fundamental equations. Rather, it draws attention to the fact that many dynamic systems are frequency dependent and yield functionally different kinds of information according to the frequency content of the input stimulus.

5. The Real Semicircular Canal

Of course, the idealized mechanical-hydrodynamic models of Figs. 3-1 and 3-3 are highly simplified compared to the anatomical structure of the real biological organ. The question arises, therefore, to what extent can the above conclusions be applied to the real end organ? Some aspects of this question have already been mentioned. Here we shall examine (a) the respective parts played by endolymph and perilymph, (b) the influence of varying canal diameter, and (c) the mechanical response of the cupula itself.

(a) Endolymph and Perilymph

Several authors have measured the electrolytic (e.g., Smith et al., 1954; Citron et al., 1956; Rauch, 1958, 1964; Rauch and Kostlin, 1958; Murray and Potts, 1961; Fernández, 1967; reviewed by Trincker, 1962, 1965, and Lowenstein, 1974) and physical (e.g., Kaieda, 1930; Rossi, 1914; von Békésy, 1960; Money et al., 1966, 1971; Steer, 1967; Steer et al., 1967; ten Kate, 1969; ten Kate et al., 1970) properties of the endolymph and perilymph, summarized results having been shown in Table 2-1 of Chapter 2. For the present purpose, note that the two fluids appear to be of slightly different density and that they are physically separate from one another.

The analysis of canal hydrodynamics outlined in previous sections assumes fluid flow within a single, unimpeded lumen of closed fluid circuit. Most authors have confined their attention to variations on this theme, assuming the relevant lumen to be that of the endolymphatic canal. Is this assumption justified? The answer is not obvious. Indeed, Rejto (1939) went so far as to suggest that in some circumstances the

perilymph may play a greater part than endolymph in generating the relevant mechanoneural transduction, a suggestion recently examined quantitatively by Anliker and van Buskirk (1971). These latter authors pointed out that if the perilymphatic duct is a patent tube and if the ampullary wall is readily deformable so that perilymph flow can cause unimpeded cupula movement, then in man "the contribution of the perilymph to the volumetric displacement of the cupula is almost 200 times that of the endolymph." Given the above assumptions, there is no reason to doubt this conclusion since it is based on sound mechanical analysis. However, over and above their stated assumptions this conclusion stems from the fact that in man the cross-sectional area of the perilymphatic duct (i.e., osseous minus endolymph canal areas) is some tenfold larger than that of the endolymphatic duct. But in cat and guinea pig, for example, the situation is quite different, the area of the perilymphatic duct occupying only about half that of the endolymphatic duct (these estimates represent very rough approximations from the precision anatomical data of Curthoys et al., 1977b).

Following this lead, recall that a study of the functional significance of canal dimensions, based on measurement of the *endolymphatic* duct (Jones and Spells, 1963), revealed a tight and systematic allometric pattern which matches well the likely patterns of head movement in man, cat, and many other animals of differing size (see Fig. 3-5). This would certainly not have been the case if that analysis had been based on measurements of the perilymphatic duct. Furthermore, as pointed out by Dohlman and Kuehn (1973), the anatomical assumptions of Anliker and van Buskirk cannot be upheld. Thus the lumen of the perilymphatic duct is not unobstructed; rather, it is crossed by numerous physical obstructions in the form of trabeculae and vessels. Furthermore, the assumption of a readily deformable ampullary wall was shown to be untenable (at least in pigeon) because of its being firmly bound to the osseous canal at its outer circumference. Hence, although it is important not to discredit the well-founded hydrodynamic implications of Anliker and van Buskirk's analysis, the weight of evidence strongly favors the common view that in response to adequate stimulation the significant mechanical response of the canal is primarily dependent on fluid circulation in the endolymphatic duct.

Given this conclusion, we are left with an uncertain mechanical function for the outer, perilymphatic fluid "bath" which encloses the "working" endolymphatic tube. Perhaps this "bath" serves to equalize pressures throughout the surface of the vulnerable inner tube, and/or to provide an electrolytic substrate for the generation of ampullary potentials contributing to mechanoneural transduction in the end organ (see Chapter 4, Section B).

(b) Nonuniform Diameter of the Endolymph Tube

The expression in equation (3.17) is derived from the idealized rigid canal of Fig. 3-3 in which the "cupula" is represented by a watertight piston whose movement is constrained within the tube by linear springs. Furthermore, both the piston and the springs are assumed to contribute no inertial or viscous forces to the system as a whole. How realistic are these assumptions?

Change of cross-sectional area could of itself introduce two main kinds of response modification. First, a transition from streamlined to turbulent fluid flow would invalidate the Poisseulle-Hagen equation used to describe the viscous moment due to fluid circulation [defined by the term B in equation (3.11)]. That this is extremely unlikely can be inferred from the fact that the (dimensionless) Reynolds number of this very small hydrodynamic system is several orders of magnitude less than that associated with the transition to turbulent flow.

A more significant source of response modification due to variation of canal diameter is the attendant change in magnitude of the term I/B in equation (3.12). Recall that this term, describing the ratio of inertial "driving" to viscous "opposing" forces, defines three functionally important parameters: (1) the sensitivity (gain) of response [equations (3.7a) and (3.7b)], (2) the rise time of the integrating response, defined by the time constant of equations (3.8) and (3.14) as well as Fig. 3-2, and (3) the upper cutoff frequency of the angular velocity mode of transduction seen in the idealized frequency response of Fig. 3-4. As seen in equation (3.12), this term is proportional to the square of the internal radius of the canal and hence must be substantially affected by this dimension.

Referring to Figs. 2-2 and 2-3, it can be seen that, as pointed out by Money et al. (1971), the thin endolymphatic tube occupies about three-fourths of the complete hydrodynamic circuit. Suppose the remaining one-fourth was occupied by an expanded tube of diameter equal to the mean diameter of the utricular sac, say roughly 0.15 cm in man. Then, for calculation of the viscous coefficient, B, in equation (3.11), we must compute two values of the net frictional force (F) acting circumferentially around the circuit, namely F_1 and F_2 for the thin and the thick portions of the tube, respectively. Assuming that the ratio of circumferential lengths occupied by the thin and thick sections is 3:1, then from equations immediately prior to equation (3.11) it can be shown that

$$\frac{F_2}{F_1} = \frac{1}{3} \cdot \frac{r_1^2}{r_2^2} \simeq \frac{1}{100} \tag{3.31}$$

where $r_1 (= 0.014$ cm) and $r_2 (= 0.08$ cm) are the internal radii of the thin

and the thick tubes, respectively. Thus the contribution of viscous forces in the thicker tube would be negligibly small and the flow pattern become dominated by the thin tube. With the thin tube occupying about three-fourths of the idealized circular torus this would yield a value of B, which is about 0.75 the idealized value.

With regard to moment of inertia (I), there has been controversy concerning the influence of variations in diameter introduced by the ampulla and the utricle. Early authors assumed that the complete endolymph ring should be taken into account but that in the widened portions of the circuit the relevant cross section should be that of the thin endolymphatic canal (von Gaede, 1922; Steinhausen, 1933; van Egmond et al., 1949). Some later authors have continued to accept this view (Fernández and Valentinuzzi, 1968; Money et al., 1971; Melvill Jones and Milsum, 1971). More recently, Oman and Young (1972) have suggested that the total mass of endolymph including that in the utricle and ampulla should be taken into account. Curthoys et al. (1977b), following this latter view, together with new data on the system's physical dimensions, arrived at considerably larger estimates of I than those previously quoted; for example, in cat, about twice the value arrived at by Fernández and Valentinuzzi (1968). The reader is referred to Table 4 on p. 29 of Curthoys et al. (1977b) for the most recent comprehensive assessment of canal constants. However, since the correct solution of the physical problem remains somewhat uncertain, we have to await experimental measurement of the actual end organ's dynamic response for a definitive statement of detailed values for a given species. Nevertheless, even with, say, a threefold increase of I and a 0.75 reduction of B, the value of $T_1 = I/B$ remains on the order of 10–20 msec. The difference is one of small magnitude rather than of kind. Hence the general conclusion that the highly simplified model of Fig. 3-3 provides a realistic representation of the general mode of the canal's hydrodynamic response still appears to hold.

(c) The Cupula in the Ampulla

Steinhausen (1931) produced a striking series of photographs of the intact but isolated pike's canal, one of which is reproduced in Fig. 2-4 of Chapter 2. The living cupula was seen to form a barrier in the ampulla extending fully across the lumen, acting like a watertight swing-door, hinged at its base, the crista ampullaris. Dohlman (1935) developed the visualization technique of Steinhausen to the point where he could make ciné film records of the cupula being "swung" from side to side by means of a micrometer-driven syringe, or by caloric stimuli, while maintaining complete physical separation of fluid on either side of the cupula. That

this separation is maintained despite relatively large forced angular displacements of the cupula is certainly evident in Dohlman's film records. A plausible schema began to emerge in which fluid circulation due to angular acceleration deflects the "swing-door" cupula through an angle in the ampulla strictly proportional to the linear displacement of fluid induced in the endolymphatic canal by change of head angular velocity.

As Steinhausen (1933) first observed, such a deflection would in all probability cause elastic distortion of tissue, and hence the presence of the cupula would introduce an elastic restoring force identified in equation (3.15) as depending on the spring coefficient, K. If this were very small, a large value of the term B/K would result, owing to the relatively large viscous term, B, due in turn to the small internal diameter of the endolymphatic duct. He pointed out that this would account for the hitherto unexplained errors of rotational sensation described in an earlier section and exemplified in Fig. 3-8.

But what degree of cupular movement is really to be expected during natural stimulation, bearing in mind the very small circulation of fluid likely to be incurred due to the physical principles examined above?

To answer this, we may note that the ratio (α) of cupula angle (ϕ) to angle of fluid displacement around the tube (θ) can be calculated from the expression (Jones and Spells, 1963)

$$\alpha = \frac{\phi}{\theta} = (\pi r)^2 R/V_a \tag{3.32}$$

where V_a is ampulla volume and r and R are as before.

Fernández and Valentinuzzi (1968) gave numerical values of $V_a = 546$ mm$^3 \times 10^{-3}$, $R = 1.7$ mm, and $r = 0.115$ mm for cat. Using these dimensions, the expression in equation (3.32) yields a value of $\alpha = 0.4$, which is close to the mean value of 0.50 estimated by Jones and Spells from 44 different mammalian species. We may now express the gain of response in terms of cupula angle per unit change of head angular velocity. Using Fernández and Valentinuzzi's (1968) data for cat, this becomes

$$\frac{\alpha I}{B} = \frac{0.4}{657} = 6.1 \times 10^{-4}$$

or using the corresponding value of $I/B = 1/150$ from Curthoys et al. (1977b)

$$\frac{\alpha I}{B} = 27 \times 10^{-4}$$

The former value of 6.1×10^{-4} implies a threshold angular displace-

ment of the cupula of around 1.2×10^{-3} deg of arc for a cat whose lowest neural threshold in a single central unit has been reported as around 2°/sec (Melvill Jones and Milsum, 1971); this value would be modified to 5.2×10^{-3} deg using the data of Curthoys et al. (1977b). Even a high head angular velocity of, say, 500°/sec would deflect the cat's cupula through only a degree or so of arc! That these numerical values are not unrealistic physiologically is suggested by the fact that similar threshold hair deflections in the otolith organ and organ of Corti may be estimated from the data of de Vries (1950) and Johnstone and Johnstone (1966), respectively.

The likelihood of very small deflections leaves us with an open question: Does the cupula move in the fashion described by Steinhausen and Dohlman in the 1930s (but more recently reinterpreted by Dohlman, 1969) or as an elastic drum attached all around its periphery (Igarashi, 1967; Landolt et al., 1975; McLaren and Hillman, 1976, 1977; Oman et al., 1978; see also Fig. 2-5)? Addressing this question, Oman and Young (1972) pointed out that because of the very small magnitude of the hydrodynamic response it may not really matter whether the cupula "swings" like a watertight door or deflects as a "bulging" diaphragm attached all around its perimeter. In either case the response would essentially correspond to that of equation (3.17), with the ratios I/B and B/K determining the sinusoidal bandwidth of angular velocity transduction due to hydrodynamic integration of the imposed angular acceleration.

(d) Linear Accelerations and the Canal

Thus far both structural and physiological evidence indicates that the normal adequate stimulus for the canal is rotational acceleration. Certainly for the rigid, circular model of Fig. 3-1 there is physical proof that linear acceleration could not generate net torque around the fluid circuit (e.g., von Gaede, 1922). Even the "cupula" model of Fig. 3-3 would be insensitive to linear acceleration if the densities of the contained "piston" and "spring" structures were the same as that of the endolymph. However, in the real semicircular canal several additional factors could be at play; for example, density differences between the cupula and endolymph and between the endolymph and perilymph.

There certainly have been neurophysiological indications in peripheral nerve–end organ preparations suggesting that changes in the prevailing direction of gravity can cause alteration of the primary afferent canal response (Ledoux, 1949; Lowenstein, 1970; Estes et al., 1975; Goldberg and Fernández, 1975b). However, as pointed out by Benson (1974), both Ledoux and Lowenstein were currently employing methods which opened the perilymphatic space, while the earlier investigator, Ross

(1936), who left the perilymphatic canal intact, did not find any effect due to gravity. Furthermore, that caloric influences introduced by open surgery can lead to artifactual gravity sensitivity was shown by Goldberg and Fernández (1975a,b), although the close proximity of single neurons with and without such sensitivity provides clear evidence that other factors are at play (Estes et al., 1975).

Could cupula-endolymph density differences contribute an effect? This seems unlikely since the results of ten separate estimates of cupula and endolymph specific gravities in pigeons revealed identical mean values of 1.0087, measured at 23.0°C relative to water at 4°C (Money et al., 1971).

The case appears to be different for endolymph and perilymph, the separate specific gravities having been estimated as 1.0087 ± 0.00049 (SD) and 1.0076 ± 0.00058 (SD), respectively, for the pigeon by Money et al. (1966). Although such a difference would not necessarily generate net torque in a steady linear accelerative field of unchanging direction, rotation of a linear acceleration vector might well do so, acting in the manner of a "roller pump" (Benson and Bodin, 1966). Using a corresponding mathematical model, Steer (1967) showed that as little as 3% change in the cross-sectional area of the duct due to this phenomenon could be expected to produce an above-threshold cupula deflection.

That this might be so is suggested by the fact that vestibular-dependent nystagmus is regularly generated in humans (Graybiel and Johnson, 1963) and cats (Correia and Money, 1968; Benson et al., 1970) by change of direction in a purely linear acceleration vector, as will be seen in later sections. Similarly, on rotation about a horizontal axis for a sufficient length of time to warrant the expectation of disappearance of any residual mechanical canal response to the rotation, there remains a more or less vigorous pattern of vestibular-dependent nystagmic eye movements in the dark (Guedry, 1965; Benson and Bodin, 1966; reviewed by Benson, 1974), presumably attributable to the gravitational vector sweeping around the rotating body (Benson and Barnes, 1973).

Taking into consideration the available facts, it seems likely that this form of "anomalous" mechanical canal stimulation probably does occur. Indeed, Benson et al. (1970) estimated that it could be responsible for as much as one-fourth to one-third of the physiological canal response to normal head rotations about a horizontal axis; in which case the effect could be functionally quite significant. The matter is complicated by Benson's (1974) failure to detect continuous "endolymph" flow in a dynamically similar (Spells, 1961) mechanical model of the combined system during its constant rotation in a vertical plane, although he was careful to point out the difficulty of ensuring realistic simulation of the living canal.

Furthermore, using an isolated elasmobranch preparation, Lowenstein (1970) could find no modulation of canal primary afferents when the system was exposed to a rotating linear acceleration vector of $0.3g$ magnitude and 31 rad/sec. The matter is obviously not a simple one and cannot be considered finally settled at the time of writing.

Summarizing, it seems unlikely that circulation of perilymph provides a significant contribution to cupular movement in the ampulla during canal rotation, leaving circulation of fluid in the endolymphatic duct as the main contributor to the canal's mechanical response. Nevertheless, differences in density between the two fluids could theoretically lead to endolymph circulation due to changes of direction of purely linear acceleration and of gravity relative to the head. Perhaps the phenomenon accounts for some of the demonstrated patterns of response in the real system to such stimuli. Other possible origins of these anomalous responses will be discussed in Chapter 4. With regard to changes of "canal" diameter, the enlarged utricular cross-section probably reduces the viscous coefficient, B, by about 25% relative to that expected from the continuous thin circular tube of Fig. 3-1. The effect of increased utricular diameter on the relevant moment of inertia (I) seems somewhat uncertain, but may increase its real value by as much as fourfold relative to that of the continuous thin, circular tube. However, even taking these two factors into account, the *mode* of expected hydrodynamic response is unlikely to be significantly altered by them over the frequency range of natural head movements, although the numerical *gain* of hydrodynamic response ($\theta/\dot{q}$) might be raised by some four- to fivefold. Even so, the expected maximum extent of cupular movement would remain on the order of 1°. The fact that such small cupular movement is to be expected permits the cupular membrane to be attached all around its periphery without modifying the essential features of the organ's hydrodynamic response. This appears to be the case in some species, although the fact remains that under artificial stimulus conditions "swing-door"-like movements of considerable magnitude have been seen to occur without destroying the watertight characteristic of the cupula in the ampulla.

C. THE OTOLITH ORGANS

1. Acceleration and Gravity

The morphology of the semicircular canals fits them for inertial transduction of movement in the three rotational degrees of freedom and

that of the otolith organs for the complementary function of inertial transduction of linear acceleration in the three translational degrees of freedom. There is, however, an important additional complication due to the Einsteinian principle that there is ambiguity between measurements of gravity and linear acceleration (e.g., Einstein, 1945; Barlow, 1964). As emphasized by Mayne (1974) and Young (1974), this fact necessitates separation of gravitoinertial information into static gravitational and dynamic linear accelerative components before there can be a useful contribution of the organ's response to either identification of attitude relative to the gravitational vertical or translational movement relative to inertial space. The uncertain question of how this occurs will be discussed later. At this stage we will examine response characteristics of the end organ itself, independently of whether they derive from forces of gravitational or accelerative origin.

2. Physical Response: Theoretical

As we have seen in Chapter 2, the mammalian utricular and saccular otolith organs comprise a ciliated sensory epithelium surmounted by a crystalline deposit (the otoconia) of calcium salts consisting mainly of calcium carbonate in calcite form (Carlström et al., 1953; Carlström, 1963), each separate otoconium constituting a crystalline organic matrix which includes a nucleus (Lim, 1973). The structures which intervene between the surface of the sensory epithelium and the otoconial mass are highlighted in the SEM microscopic photographs of Figs. 2-11, 2-12, 2-14, and 2-17. It seems that the bundles of cilia are fitted into the otolith membrane with the otoconia being bound over the surface of this membrane by a thin cobweblike net (Fig. 2-13).

Histological and microanatomical preparations thus suggest a simplified functional arrangement diagrammed in Fig. 3-9. Linear acceleration of the whole system to the right ($\ddot{x}$) causes the otoconial mass to be "left behind" by virtue of its inertia, thereby bending the cilia of the sensory haircells and so initiating the process of mechanoneural transduction described in the next chapter. Note that since the otoconial mass gets "left behind," its actual acceleration relative to space ($\ddot{y}$) will be less than that of the whole system.

Figure 3-9B reduces this conceptual scheme to an idealized mechanical model suitable for developing a mathematical description of the system's likely behavior. In this idealized system a spherical body of mass M and density ρ_m is centralized between springs in a cylindrical tube of larger internal diameter than the body and containing a fluid (endolymph)

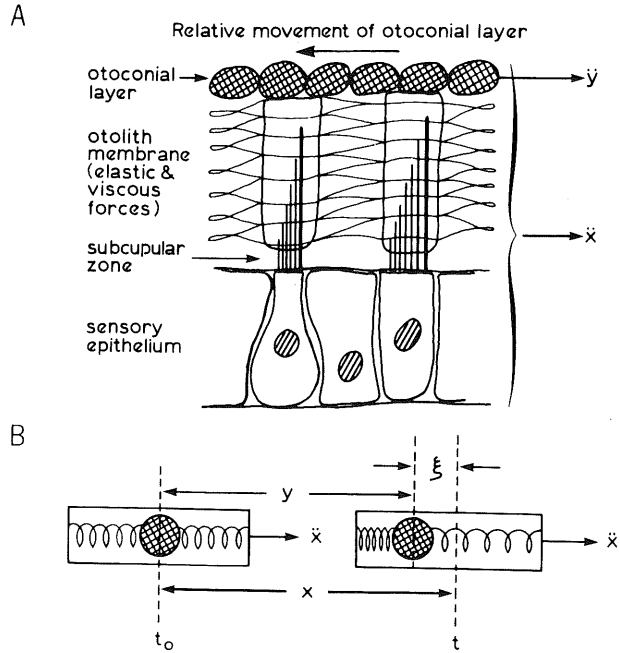

Fig. 3-9. A: Presumed mechanical linkages between the macular epithelial cells and overlying structures. B: Elementary mechanical analogy. $\ddot{x}$, Linear acceleration of head relative to space; $\ddot{y}$, acceleration of otoconia relative to space; ξ, relative displacement of otoconia and macula; t_0, position of end organ at commencement of acceleration; t, position after time t.

of density ρ_e. Analogous with the canals, linear acceleration ($\ddot{x}$ in Fig. 3-9B) sets up a system of forces acting on the otoconial body such that the sum of leftward-acting forces equates to the sum of those acting to the right.

Acting to left is the inertial force, $M\ddot{y}$, where $\ddot{y}$ is the acceleration of the body's mass, M, relative to inertial space. The resulting displacement of this mass *relative to the tube* ($\xi = x - y$) will generate opposing viscous ($B'\dot{\xi}$), and spring ($K'\xi$) forces acting to the right. As with the canal, the velocity proportionality constant associated with viscous drag is justified on the assumption of strictly laminar flow due to the small size of the labyrinthine system. In this case, however, since the otoconial mass is immersed in endolymph fluid of density ρ_e, any acceleration, whether due to linear movement or to gravity, will generate a buoyancy force acting according to Archimedes' principle in the direction of imposed acceleration and equal to $(\rho_e/\rho_m)\, M\ddot{x}$, where ρ_m is the density of the otoconial mass.

Equating the specific force acting to the left and the sum of all other forces acting to the right [cf. equations (3.15) and (3.16) for the canal]

$$M\ddot{y} = \frac{\rho_e}{\rho_m} M\ddot{x} + B'\dot{\xi} + K'\xi$$

or, since $y = (x - \xi)$,

$$\left(1 - \frac{\rho_e}{\rho_m}\right)\ddot{x} = \ddot{\xi} + \frac{B'}{M}\dot{\xi} + \frac{K'}{M}\xi$$

Although very similar to the canal equation (3.16), an important difference is seen in the term $(1 - \rho_e/\rho_m)$, which would become zero if the density of endolymph and the otoconia were identical. The system functions only insofar as these two densities differ from one another. Numerically $\rho_e = 1.01$ (Money et al., 1971) and $\rho_m = 2.71$ (Carlström et al., 1953) so that a significant mechanical response is obtained. Note incidentally that the value of M must be increased, probably by a factor of about 1.3, to account for inertial forces due to acceleration of local fluid displaced by relative movement of the "otoconial" body (de Vries, 1950). Thus, assuming the equivalent mass $m = 1.3M$, the above equation becomes

$$\left(1 - \frac{\rho_e}{\rho_m}\right)\ddot{x} = \ddot{\xi} + \frac{B'}{m}\dot{\xi} + \frac{K'}{m}\xi \tag{3.33}$$

where, summarizing the symbols,

$\ddot{x}$ = linear acceleration of the whole system
B' = viscous drag coefficient
K' = spring coefficient
m = equivalent mass of otoconial body
ξ = *relative* displacement of the otoconial body (i.e., the response of the sensor)
ρ_e = density of endolymph
ρ_m = density of the otoconial body

The corresponding Laplace transfer function is given by

$$\frac{\xi}{\ddot{x}}(s) = \left(1 - \frac{\rho_e}{\rho_m}\right) \cdot \frac{1}{s^2 + (B'/m)s + K'/m} \tag{3.34}$$

$$= \left(1 - \frac{\rho_e}{\rho_m}\right) \cdot \frac{m/K'}{(T_1 s + 1)(T_2 s + 1)} \tag{3.35}$$

where $T_1T_2 = m/K'$ and $T_1 + T_2 = B'/K'$.

However, whereas the simplicity of the canal's elementary mechanics permits numerical estimation of relevant parameters and hence analytical prediction of likely patterns of dynamic response, this is not possible for the complex mechanical system of the otolith organ. On the one hand, the "effective" mass of the compound otoconia–otolith membrane–cilia assembly is not readily assessed. On the other hand, the irregular surface area in the interstices of the otolith membrane meshwork and the quite unknown frictional forces likely to arise within it deny the possibility of calculating the viscous coefficient B'. Consequently, we must turn to the experimental, rather than analytical, approach for further understanding of the end organ response.

3. Physical Response: Experimental

The majority of mammalian experimental information derives from studies of neural, reflex, and perceptual response which will receive more detailed treatment later.

There are, however, some relevant results available from direct observation of the end organ, although these appear to have been confined to lower animals.

De Vries' (1950) measurement of otolith membrane movement in fish by X-ray photography still remains the major source of direct information, although the somewhat similar experiments of Vilstrup and Vilstrup (1952) provide supporting evidence as do the earlier attempts at micromechanical stimulation of the end organ (Ulrich, 1934/1935; Steinhausen, 1934/1935). Figure 3-10 plots the measured movement of the saccular otolith in a 24-g ruff fish. The displacements are of course small, but well within the resolution of measurement (quoted as about 0.005 mm), as may be seen by the tight clustering of points about a straight line between the limits of $\pm$ 1g. Variation of the components of applied acceleration in the

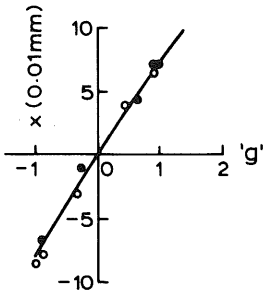

Fig. 3-10. Measured movement of saccular otolith in the ruff fish. (de Vries, 1950.)

plane of the saccular macula was obtained by rotating the animal from a head-up to a head-down attitude. All responses in this part of the curve were obtained in a static condition and hence represent the static mechanical response. A single value of 23×0.01 mm was obtained at $11g$ (not shown here) by generating a centripetal acceleration by means of rotation of the whole fish at the phenomenal angular velocity of 8.3 rev/sec. However, even this estimate appears to have been reliable since no measurable movement of the skull occurred relative to a fixed frame of cross-wires. Evidently the static mechanical response was closely linear over the range $\pm 1g$, but the system saturated at some higher level.

The author ambitiously extended his observations to dynamic stimuli in the form of oscillations (vibrations) "produced by connecting the fish holder plus the photographic plate to an eccentric on the axis of a motor." From these results and measurements of otoconial mass and volume, de Vries attempted to derive values for the constants describing equation (3.33). These values naturally varied from one animal to another according to size and species. The general conclusion was that the system is close to being critically damped, so that it does not oscillate after a transient disturbance although it exhibits a rapid response. De Vries (1950) concluded that the natural frequency (N_0) of the system could be described by the relation

$$N_0(\text{Hz}) = \frac{1}{2\pi}\sqrt{g/s} \tag{3.36}$$

where g is gravitational acceleration and s is the peak-to-peak displacement of the oscillatory response. With s on the order of 0.2 mm, this expression yields a value of N_0 around 40 Hz, which presumably would provide an adequate frequency response for reliable registration of accelerations during natural movement. The smaller values of s found in smaller animals would lead to a higher frequency response which would appropriately cater to the more rapid movements to be expected from smaller animals, although quantitative allometric studies on this feature do not appear to have been reported.

The direct nature of de Vries' measurements permits a rough comparison of mechanical sensitivity with canal and cochlea end organs. Thus the mean of eight measurements of saccular otolith displacement on turning the fish through 180° (a peak-to-peak stimulus of $2g$) was 0.065 mm, which corresponds to 33 μm/g. Insofar as threshold sensitivity for saccular stimulation in man (Jongkees and Klijn, 1959; Meiry, 1966; Melvill Jones and Young, 1978) and cat (Daunton and Melvill Jones, 1973) appears to be around $0.005g$, this suggests a threshold linear otolith displacement of around 0.15 μm. Assuming a distance of about 35 μm (e.g.,

Igarashi and Kanda, 1969; Igarashi and Alford, 1969) between the otoconial layer and sensory epithelium as well as a simple mechanical pivot of the cilia at the epithelial surface, this in turn yields an estimate of threshold deviation of cilia relative to the epithelial surface of around ¼ deg of arc. De Vries points out that this is relatively large and that utricular otolith measurements were considerably less than this, being below the resolution of his method for simple gravitational stimuli. With a quoted resolution of 5 μm, this indicates a threshold angular displacement of cilia less than 2.0×10^{-2} deg, although how much less we cannot say.

The matter is complicated by uncertainty as to exactly how otolith movement "bends" the hairs and perhaps de Vries' attempt to estimate the energy absorbed at threshold stimulation is more realistic. His conclusion was that the minimum perceptible energy for threshold utricular stimulation is about 2.1×10^{-11} erg and that this is similar to the corresponding value for a semicircular canal. The dynamic characteristics of otolith response have been investigated by several authors using neural, reflex, and perceptual indicators of response. Although treated more extensively in later chapters, salient features are mentioned here for comparative purposes.

Using perceptual criteria, Meiry (1966) and Young et al. (1966) arrived at an empirical transfer function of the form

$$\frac{\text{perceived response}}{\ddot{x}}(s) = \frac{m/K'}{(T_1 s + 1)(T_2 s + 1)} \tag{3.37}$$

where $T_1 = 10$ sec and $T_2 = 0.66$ sec. The analogy with equation (3.35) is obvious and suggests that the perceived response provides a close analogue of physical otolith displacement. According to this conclusion, instead of the critically damped system of de Vries we would have an overdamped system leading to a frequency response reminiscent of that of the canal but having relatively less viscous damping. The low value of the upper cutoff frequency [0.25 Hz = $1/(2\pi T_2)$] is, however, very far from de Vries' estimated natural frequency of around 40 Hz. One may guess that, as with the canals (van Egmond et al., 1949), the rapid mechanical response of the end organs might well be masked by relatively "sluggish" perceptual estimates of response dynamics. Incidentally, equation (3.37) was later modified by Young and Meiry (1968) to account for interaction of adaptive phenomena and a residual static response. Since these attributes were presumed to have a neural origin, the implications will not be pursued further here.

Mayne (1974) adopted another approach based on the fundamental premise emphasized earlier that it should be possible to separate static gravitational from dynamic linear accelerative influences. Basing his ar-

guments on the neural data of Lowenstein and Roberts (1949), he formulated the relation

$$\frac{\xi}{\ddot{x}}(s) = K_1 - K_2 \frac{1}{(T_1 s + 1)(T_2 s + 1)} \tag{3.38}$$

The functional interpretation of this expression is that it yields the characteristics both of a conventional dc linear accelerometer (with the first right-hand term, K_1, as the proportionality constant) and of a differentiating accelerometer (second right-hand term). This would permit separation of short-duration transient accelerations, specifically by virtue of the time constant, T_1, determining the critical cutoff or break frequency.

More recently a somewhat more elaborate dynamic response analysis has been undertaken experimentally by Fernández and Goldberg (1976c), based on a careful study of response patterns in monkey primary afferent neurons identified as originating from otolith end organs. Their neural results will be detailed in Chapter 4. However, at this stage we may note that they identify a "first-order lag operator," $H_m = 1/(T_m s + 1)$, which "may reflect the mechanics of otolith motion." The median value calculated for the time constant T_m was 16 msec, which, as could be expected at the level of the first-order neuron, is much shorter than the somewhat analogous value of 0.66 sec arrived at from the perceptual data of Young and Meiry quoted above.

In summary, the biophysics of the otolith end organs is much less well researched than that of the canals. However, it is now well established that the basic nature of otolith reaction to its adequate stimulation is an inertia-generated shearing movement of the otoconial layer parallel to the underlying surface of the sensory epithelium. It should be recorded that there have been serious doubts, and indeed contrary views, expressed by authoritative writers in the earlier literature (e.g., Magnus, 1924; Quix, 1923). But today this conflict can be considered settled in view of direct observations by X-rays as described above (de Vries, 1950; Vilstrup and Vilstrup, 1952), measurement of physiological response to the application of highly localized forces to the end organ by means of hairlike probes (Steinhausen, 1934/1935; Ulrich, 1934/1935; Szentágothai, 1952), as well as the abundance of evidence from neural (e.g., Lowenstein and Roberts, 1949; Fernández and Goldberg, 1976a) and behavioral (e.g., von Holst, 1950a,b; Schöne, 1964, 1971; Jongkees, 1967) patterns of response to adequate stimulation.